Vivek Ruhela

Postdoctoral Research Scientist, Columbia University Irving Medical Center

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Research Profile

Computational genomicist developing ancestry-aware statistical and AI-driven methods to identify molecular mechanisms of Alzheimer's disease and other complex disorders. My work integrates whole-genome sequencing, brain RNA-seq, runs of homozygosity (ROH), GWAS, eQTL mapping, fine-mapping, colocalization, and reproducible HPC workflows across diverse populations. My emerging independent research program focuses on interpretable multi-omics methods and open computational resources that improve biological discovery and generalizability across ancestries.

Academic Appointment

May 2024 - Present — **Postdoctoral Research Scientist**
Department of Neurology; Taub Institute for Research on Alzheimer's Disease and the Aging Brain; Gertrude H. Sergievsky Center; Columbia University Irving Medical Center, New York, NY, USA
Research focus: genetic fine-mapping and multi-omics analysis of Alzheimer's disease across diverse populations. Develop ancestry-aware pipelines for ROH, GWAS/eQTL integration, colocalization, meta-analysis, transcriptomics, and scalable HPC analysis.

Education

2018 - 2024 — **Ph.D., Computational Biology**
Indraprastha Institute of Information Technology Delhi (IIIT-Delhi), India
Research area: Cancer genomics.
Thesis: Design and development of AI-based computational tools for identifying predictive biomarkers and signaling pathways for blood cancer.
Advisors: Prof. Anubha Gupta (IIIT-Delhi) and Prof. Ritu Gupta (AIIMS, Delhi).

2014 - 2016 — **M.Tech., Instrumentation & Control**
ZHCET, Aligarh Muslim University, India
CPI: 9.0 (First Class Honours). Thesis: Detection of brain tumor in brain MRI. Advisor: Prof. Yusuf Uzzaman Khan.

2006 - 2010 — **B.Tech., Electrical Engineering**
BSACET College, Dr. A.P.J. Abdul Kalam Technical University (AKTU), India

Selected Oral Presentations & Research Talks

2026 — **Oral Research Talk**
National Institute on Aging (NIA), National Institutes of Health (NIH), Rockville, MD, USA
Maternal-Offspring Haplotype Transmission Reveals AD-Relevant Signals in ADSP Dyads.

2026 — **Oral Presentation - Abstract ID 2701**
AD/PD 2026 International Conference on Alzheimer's and Parkinson's Diseases and Related Neurological Disorders, Copenhagen, Denmark
Maternal-Offspring Haplotype Transmission Reveals AD-Relevant Signals in ADSP Dyads.

Research Experience

2018 - 2024 — **Doctoral Researcher**
Indraprastha Institute of Information Technology Delhi (IIIT-Delhi), India
Developed AI-driven biomarker discovery and genomic modeling approaches for multiple myeloma.
Developed miRPipe, miRSim, BDL-SP/Bio-DGI and related computational frameworks for small-RNA sequencing, disease-pathway modeling, and multi-omics biomarker discovery.

2015 - 2016 — **Graduate Researcher (M.Tech. Thesis)**
Aligarh Muslim University, Aligarh, India
Applied image processing and machine-learning methods to brain-tumor detection in MRI using MATLAB, SVMs, and feature extraction.

Research Publications

Peer-reviewed journal articles

1. **Ruhela, Vivek**, Gupta, R., Oberoi, R., & Gupta, A. (2025). A comprehensive targeted panel of 295 genes: Unveiling key disease initiating and transformative biomarkers in multiple myeloma. *Computers in Biology and Medicine, 196, 110619.* doi:10.1016/j.compbiomed.2025.110619
2. **Ruhela, Vivek**, Jena, L., Kaur, G., Gupta, R., & Gupta, A. (2023). BDL-SP: A bio-inspired DL model for the identification of altered signaling pathways in multiple myeloma using WES data. *American Journal of Cancer Research, 13(4), 1155.* PMC
3. Farswan, A., Gupta, A., Jena, L., **Ruhela, Vivek**, Kaur, G., & Gupta, R. (2022). Characterizing the mutational landscape of MM and its precursor MGUS. *American Journal of Cancer Research, 12(4), 1919.* PubMed
4. **Ruhela, Vivek**, Gupta, A., Sriram, K., Ahuja, G., Kaur, G., & Gupta, R. (2022). A unified computational framework for a robust, reliable, and reproducible identification of novel miRNAs from RNA sequencing data. *Frontiers in Bioinformatics, 2, 842051.* doi:10.3389/fbinf.2022.842051
5. Kaur, G., **Ruhela, Vivek**, Rani, L., Gupta, A., Sriram, K., Gogia, A., et al. (2020). RNA-seq profiling of deregulated miRs in CLL and their impact on clinical outcome. *Blood Cancer Journal, 10(1), 1-9.* doi:10.1038/s41408-019-0272-y

Preprint

1. Cieza, B., Pandey, N., **Ruhela, Vivek**, Ali, S., & Tosto, G. (2025). SAGA (Simplified Association Genomewide Analyses): A user-friendly pipeline to democratize genome-wide association studies. bioRxiv. doi:10.1101/2025.08.25.672146

Conference/supplement publication

1. **Ruhela, Vivek**, Farswan, A., Gupta, A., Sriram, K., Kaur, G., & Gupta, R. (2021). P-035: AI-based models for the identification of critical genetic biomarkers to distinguish MM from MGUS using WES data. *Clinical Lymphoma Myeloma and Leukemia, 21, S57*. doi:10.1016/S2152-2650(21)02169-8

Research Software & Open Resources

- **BRAINscape (2025)** - R/Shiny multi-omics exploration platform for MU-BRAIN, supporting differential-expression and cis/trans-eQTL interrogation with ancestry-stratified visualization.
- **SAGA (2025)** - User-friendly, reproducible workflow for genome-wide association analysis.
- **miRPipe (2022)** - Reproducible framework for identification of novel miRNAs from RNA-sequencing data.
- **miRSim (2021)** - Seed-based synthetic small non-coding RNA sequence simulator.
- **BDL-SP / Bio-DGI** - Bio-inspired deep-learning and graph-based frameworks for disease-pathway and biomarker discovery in multiple myeloma.

Selected Conference Abstracts & Presentations

2025	Vivek Ruhela, Giuseppe Tosto, Joseph L. Jacobson, Ernesta Meintjes, Sandra W. Jacobson, R. & Colin Carter. Gene-Based Genome-Wide Approaches Integrating Maternal and Child Genetic Data Identify Novel Risk- Modifying Genetic Loci for Fetal Alcohol Spectrum Disorders. ESBRA Conference.
2025	Ruhela, V., Cieza, B., Mayeux, R., Reyes-Dumeyer, D., Teich, A. F., & Tosto, G. Genetic colocalization of eQTL mapping and GWAS in a multiethnic brain bank: ancestry-specific regulatory architecture in Alzheimer's disease. Alzheimer's Association International Conference (AAIC).
2025	Ruhela, V., Pandey, N., Cieza, B., Barral, S., Samper-Ternent, R., Montesinos, R., et al., & Tosto, G. Fine mapping of regions of homozygosity and their role in Alzheimer's disease: insights from Peruvian and Mexican populations. AD/PD Conference.
2024	Ruhela, V., Yang, Z., Jacobson, S. W., Jacobson, J. L., Meintjes, E. M., Tosto, G., & Carter, R. C. Gene by prenatal alcohol exposure interaction effects on growth and cognition in mother-child dyads in a South African birth cohort. American Society of Human Genetics (ASHG).

Teaching Experience

2017	**Assistant Professor** Electrical and Electronics Engineering, PSIT, Kanpur, India
2016	**Assistant Professor** Electrical Engineering, Sanskriti University, India
2012 - 2013	**Course Instructor / Lecturer (Ad hoc)** ADKMP College and BSA College, India
2010 - 2012	**Lecturer** Electrical Engineering, Mangalayatan University, India

Awards, Honors & Certifications

2010	**Competition Prize for Outstanding Performance in MATLAB Code Challenge** BSA Engineering College, India
2017	**NPTEL Elite Certification** Medical Image Analysis
2017	**NPTEL Elite Certification** Enhancing Soft Skills and Personality

Technical Expertise

Computational genomics: GWAS, ROH, eQTL, fine-mapping, colocalization, ancestry-aware analysis, transcriptomics/RNA-seq, multi-omics integration.
Programming / HPC: R, Python, Bash, AWK, MATLAB, LaTeX, SLURM/HPC workflows, reproducible pipeline development.
Methods: Statistical genetics, machine learning, deep learning, graph-based modeling.

References

Available on request.